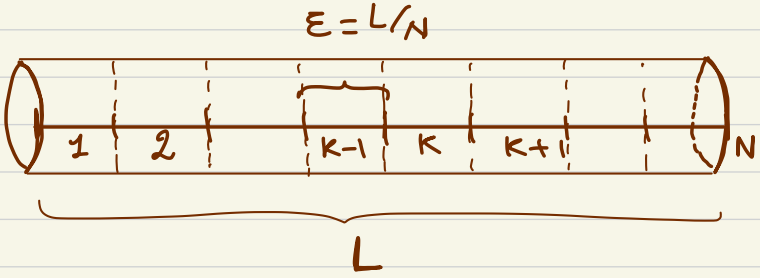
Derivation of the Diffusion equation with Fickian law

The aim of these notes is to derive the diffusion equation using the technique of discretization

In order to understand the phenomenon, we can immagine the diffusion process of a drop of milk in the coffe or the ink into the water.

We assume to have a tube filled with a fluid, having radius r and lenght L and divided in N parts of lenght $\varepsilon = L/N$. The single part is distinguished with index k.



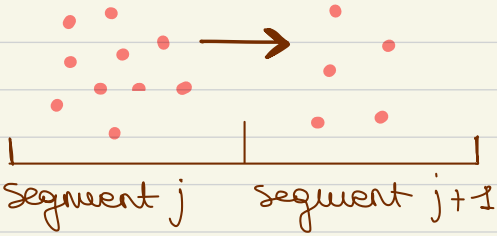
we assume the heigh of the tube to be negligible so that we consider a 1 dimensional setting.

We denote $u_k(t)$ the average concentration of the solute in the section k at time t and we proceed as if $u_k(t)$ is the concentration at any point of the segment k (the smaller ε is, the smaller the approximation's error is).

The section K is in contact with the section $K-1$, for $K \neq 1$, and with $K+1$, for $K \neq N$. Therefore, at the interface of the consecutive sections, we assume the so called **Fickian law**.

↓

the flux of the substance goes from region of high concentration to region of low concentration



Thus, we assume that the intensity of the flux is proportional to the gradient of the concentration.
Therefore,

* at the interface between the section K and $K+1$, the flux (entering in K) will be proportional to

$$[u_{K+1}(t) - u_K(t)] \cdot \frac{\nu}{\varepsilon}$$

* at the interface between the section K and $K-1$, the flux (entering in K) will be proportional to

$$[u_{K-1}(t) - u_K(t)] \cdot \frac{\nu}{\varepsilon}.$$

Therefore, denoting $S_K(t)$ the quantity of the substance in the section K at time t, we have

$$\frac{d}{dt} S_K(t) = \frac{\nu}{\varepsilon}\left[u_{K+1}(t) - u_K(t) + u_{K-1}(t) - u_K(t)\right]$$
$$= \frac{\nu}{\varepsilon}\left[u_{K+1}(t) - 2u_K(t) + u_{K-1}(t)\right]. \quad (1)$$

On the other hand, the concentration of the substance in the segment K is equal to the quantity of the substance, divided by the volume V, that is

$$u_K(t) = \frac{S_K(t)}{V} =: \underbrace{c}_{\text{constant of proportionality}} S_K(t). \quad (2)$$

Thus, from (2) we have $S_K(t) = c^{-1} u_K(t)$ and plugging the obtained identity into (1), we end up with,

$$\frac{d}{dt} u_K(t) = c\frac{\nu}{\varepsilon}\left[u_{K+1}(t) - 2u_K(t) + u_{K-1}(t)\right]. \quad (3)$$

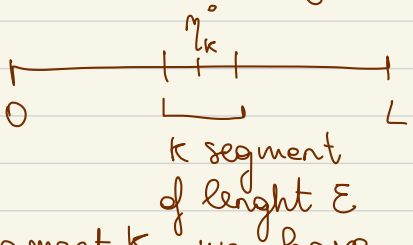
GOAL: $\varepsilon \to 0$

So far, we consider a discretized contraction. Now, we want to pass from a discrete to a continous setting, by taking the limit as $\varepsilon \to 0$

Let consider x_k° the average point in the segment k,

$$x_k^\circ = k\varepsilon - \frac{1}{2}\varepsilon = \left(k - \frac{1}{2}\right)\frac{L}{N}$$

$\varepsilon = \frac{L}{N}$



then, for any point x in the segment k, we have

$$|x - x_k^\circ| \leq \frac{\varepsilon}{2}.$$

Let now consider $\rho(x,t)$ the concentration of the substance at the point x, at time t and assume that ρ is enough regular. Then, it's easy to prove (Taylor expansion) that when ε is small, the average concentration $u_k(t)$ is approximated with an error $O(\varepsilon)$ by $\rho(x,t)$, i.e.

$$u_k(t) = \rho(x,t), \quad \forall\, x \text{ s.t. } |x - x_k^\circ| \leq \frac{\varepsilon}{2} \tag{4}$$

and

$$u_{k\pm1}(t) = \rho\left(x \pm \frac{\varepsilon}{2}, t\right). \tag{5}$$

Therefore, using (4), (5) in (3) we obtain

$$\partial_t \rho(x,t) = \frac{c\nu}{\varepsilon}\left[\rho\left(x+\frac{\varepsilon}{2},t\right) - 2\rho(x,t) + \rho\left(x-\frac{\varepsilon}{2},t\right)\right]. \tag{6}$$

However, using the Taylor expansion for any fixed t, up to the 2nd order, we get

$$\rho\left(x \pm \frac{\varepsilon}{2}, t\right) = \rho(x,t) \pm \frac{\varepsilon}{2}\partial_x \rho(x,t) + \frac{\varepsilon^2}{8}\partial_{xx}^2 \rho(x,t). \tag{7}$$

Therefore, plugging (7) into (6) we have

$$\partial_t p(x,t) = \frac{c\nu}{\varepsilon}\Big[p(x,t) + \frac{\varepsilon}{2}\partial_x p(x,t) + \frac{\varepsilon^2}{8}\partial_{xx}^2 p(x,t) + \\ - 2p(x,t) + p(x,t) - \frac{\varepsilon}{2}\partial_x p(x,t) + \\ + \frac{\varepsilon^2}{8}\partial_{xx}^2 p(x,t)\Big]$$

$$= \frac{c\nu}{4}\varepsilon^2 \partial_{xx} p(x,t).$$

Recalling that $V = \pi r^2 \varepsilon \Rightarrow c = (\pi r^2 \varepsilon)^{-1}$, thus

$$\partial_t p(x,t) = \frac{\nu}{4\pi r^2}\partial_{xx}^2 p(x,t)$$ that does not depend on ε.

Denoting by $\alpha := \frac{\nu}{4\pi r^2}$ the diffusion coefficient then we obtain the diffusion equation

$$\boxed{\partial_t p(x,t) = \alpha\, \partial_{xx}^2 p(x,t)}$$